

Experiment analysis step 5: precision comparison

Realized precision for crossover and first-period benchmark designs

W241 Plank Study

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0.1 Purpose

This document compares the realized precision of:

- the primary crossover design
- the first-period-only parallel benchmark

The comparison is based on standard errors, 95% CI widths, minimum detectable effects, and the benchmark sample size required to match crossover precision. This is more informative than retrospective observed power.

0.2 Load datasets

```
paired <- sanitize_dt_ascii(fread(derived_path("paired_analysis.csv")))
long_dt <- sanitize_dt_ascii(fread(derived_path("long_repeated_measures.csv")))

bench <- long_dt[
  in_first_period_benchmark == 1L & period == 1L
]

stopifnot(nrow(paired) == 38L)
stopifnot(nrow(bench) == 46L)
```

0.3 Precision calculations

```
alpha_two_sided <- 0.05
target_power <- 0.80
z_alpha <- qnorm(1 - alpha_two_sided / 2)
z_power <- qnorm(target_power)
mde_multiplier <- z_alpha + z_power
```

```

# Crossover / paired design
n_pair <- nrow(paired)
sd_diff <- sd(paired$diff_high_minus_low)
se_pair <- sd_diff / sqrt(n_pair)
ci_width_pair <- 2 * z_alpha * se_pair
mde_pair <- mde_multiplier * se_pair

# First-period parallel benchmark
n_high <- bench[high_tempo == 1L, .N]
n_low <- bench[high_tempo == 0L, .N]
sd_high <- bench[high_tempo == 1L, sd(outcome_tte)]
sd_low <- bench[high_tempo == 0L, sd(outcome_tte)]
se_parallel <- sqrt((sd_high^2 / n_high) + (sd_low^2 / n_low))
ci_width_parallel <- 2 * z_alpha * se_parallel
mde_parallel <- mde_multiplier * se_parallel

sd_pooled <- sqrt(
  (((n_high - 1) * sd_high^2) + ((n_low - 1) * sd_low^2)) /
  (n_high + n_low - 2)
)

# Equal-allocation parallel design needed to match the crossover SE
n_per_arm_to_match <- ceiling((2 * sd_pooled^2) / (se_pair^2))
total_n_to_match <- 2 * n_per_arm_to_match

precision_tab <- data.table(
  design = c("Primary crossover", "First-period benchmark"),
  independent_n = c(n_pair, n_high + n_low),
  observations = c(2 * n_pair, n_high + n_low),
  relevant_sd = c(sd_diff, sd_pooled),
  standard_error = c(se_pair, se_parallel),
  ci_width_95 = c(ci_width_pair, ci_width_parallel),
  mde_80_power = c(mde_pair, mde_parallel)
)

precision_tab[, `:=`(
  relevant_sd = round(relevant_sd, 4),
  standard_error = round(standard_error, 4),
  ci_width_95 = round(ci_width_95, 4),
  mde_80_power = round(mde_80_power, 4)
)]

```

0.4 Precision table

```
kable(precision_tab, caption = "Realized precision by design")
```

Table 1: Realized precision by design

design	independent_n	observations	relevant_sd	standard_error	ci_width_95	mde_80_power
Primary crossover	38	76	28.9211	4.6916	18.3908	13.1440
First-period benchmark	46	46	47.0386	13.8709	54.3730	38.8606

```

matching_tab <- data.table(
  quantity = c(
    "Observed benchmark total N",
    "Required benchmark total N to match crossover SE",
    "Required benchmark N per arm to match crossover SE",
    "Extra independent participants needed"
  ),
  value = c(
    n_high + n_low,
    total_n_to_match,
    n_per_arm_to_match,
    total_n_to_match - (n_high + n_low)
  )
)

kable(matching_tab, caption = "Parallel-design sample size needed to match crossover precision")

```

Table 2: Parallel-design sample size needed to match crossover precision

quantity	value
Observed benchmark total N	46
Required benchmark total N to match crossover SE	404
Required benchmark N per arm to match crossover SE	202
Extra independent participants needed	358

0.5 Precision comparison figure

```

plot_precision <- melt(
  precision_tab[, .(design, standard_error, ci_width_95, mde_80_power)],
  id.vars = "design",
  variable.name = "metric",
  value.name = "value"
)

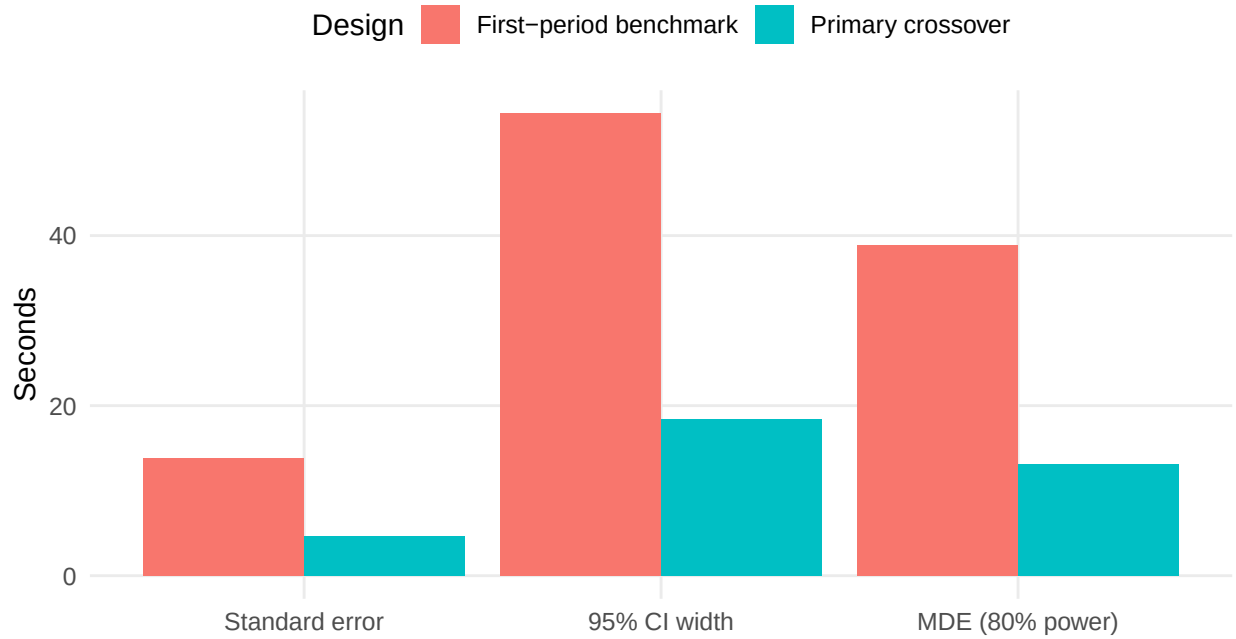
plot_precision[, metric := factor(
  metric,
  levels = c("standard_error", "ci_width_95", "mde_80_power"),
  labels = c("Standard error", "95% CI width", "MDE (80% power)")
)]

ggplot(plot_precision, aes(x = metric, y = value, fill = design)) +
  geom_col(position = "dodge") +
  labs(
    title = "Realized precision comparison",
    subtitle = "Smaller values indicate tighter design precision",
    x = NULL,
    y = "Seconds",
    fill = "Design"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid.minor = element_blank(), legend.position = "top")

```

Realized precision comparison

Smaller values indicate tighter design precision



0.6 Interpretation notes

```
precision_ratio <- se_parallel / se_pair
ci_ratio <- ci_width_parallel / ci_width_pair

interpretation_tab <- data.table(
  quantity = c(
    "Parallel SE / crossover SE",
    "Parallel CI width / crossover CI width"
  ),
  value = round(c(precision_ratio, ci_ratio), 4)
)

kable(interpretation_tab, caption = "Relative precision loss in the benchmark design")
```

Table 3: Relative precision loss in the benchmark design

quantity	value
Parallel SE / crossover SE	2.9565
Parallel CI width / crossover CI width	2.9565

0.7 Verification

```
stopifnot(n_pair == 38L)
stopifnot((n_high + n_low) == 46L)
stopifnot(se_pair > 0)
stopifnot(se_parallel > 0)
```

```
stopifnot(total_n_to_match >= (n_high + n_low))

message("Crossover SE: ", round(se_pair, 3))
## Crossover SE: 4.692
message("Benchmark SE: ", round(se_parallel, 3))
## Benchmark SE: 13.871
message("Required parallel total N to match crossover SE: ", total_n_to_match)
## Required parallel total N to match crossover SE: 404
```